

Evolution of defect states in C₂H₂-doped GaN:C studied by optical deep-level transient spectroscopy

Cite as: J. Appl. Phys. **139**, 185701 (2026); doi: [10.1063/5.0331068](https://doi.org/10.1063/5.0331068)

Submitted: 24 February 2026 · Accepted: 16 April 2026 ·

Published Online: 11 May 2026



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ABSTRACT

Carbon-doped GaN (GaN:C) is widely used as a semi-insulating buffer layer in GaN-based power devices. Although the electrical properties of GaN:C grown by carbon autodoping have been extensively studied, the influence of the carbon incorporation method remains unclear. In this study, we investigate whether the conventional assumption of discrete, quasi-isolated carbon-related deep levels remains sufficient to describe defect states when carbon is introduced via an external hydrocarbon source. GaN:C layers grown using C₂H₂ gas were characterized by current–voltage (I–V), capacitance–voltage (C–V), and optical deep-level transient spectroscopy (ODLTS). The samples exhibit a resistive and effectively fully depleted state at high carbon concentrations. ODLTS measurements reveal an evolution of trap responses from quasi-isolated states at $\sim 10^{17} \text{ cm}^{-3}$ to broad and distributed responses at $\sim 10^{19} \text{ cm}^{-3}$. These results indicate that the carbon incorporation pathway influences the nature of electrically active defect states. By comparing our results with previous reports on autodoped GaN:C, this study clarifies the distinct evolution of defect states induced by C₂H₂ doping. These findings provide insight into the realization of stable semi-insulating GaN buffer layers for power device applications.

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I. INTRODUCTION

Gallium nitride (GaN) has attracted considerable attention as a key material for next-generation power electronic devices because of its wide bandgap of approximately 3.4 eV, high critical electric field, high electron mobility, and excellent thermal and chemical stability.^{1–3} These material properties enable GaN-based devices to operate under high-voltage, high-frequency, and high-temperature conditions, thereby contributing to improved power efficiency and system miniaturization. In addition, the high saturation electron velocity and low intrinsic carrier concentration of GaN are advantageous for reducing conduction and switching losses in power devices.⁴ Owing to these characteristics, GaN has become one of the most promising wide-bandgap semiconductors for power electronics applications. In practical GaN-based device structures, such as AlGaIn/GaN high-electron-mobility transistors

(HEMTs), the properties of the buffer layer play a crucial role in determining device performance and reliability.^{5,6} In particular, a semi-insulating GaN buffer layer is required to suppress buffer leakage currents, improve breakdown voltage, and mitigate short-channel effects. To achieve high resistivity in GaN buffer layers, intentional carbon (C) doping has been widely employed.^{7,8} Carbon is known to introduce deep acceptor-like states that compensate residual n-type carriers originating from unintentional impurities, such as oxygen and silicon.^{8,9} As a result, carbon-doped GaN (GaN:C) has become a standard buffer material in commercial GaN power devices. At the same time, carbon incorporation into GaN inevitably introduces deep-level defects, which can act as carrier trapping centers.^{10–12} These traps have been linked to various reliability issues in GaN devices, including dynamic on-resistance degradation, current collapse, and persistent photoconductivity.^{13–15} Consequently, understanding the

nature and behavior of carbon-related deep levels is essential for optimizing GaN buffer layers and improving long-term device stability. Under high electric field conditions in AlGaIn/GaN HEMTs, distributed or interacting trap states in the buffer layer can dynamically capture and emit carriers, leading to current collapse and dynamic on-resistance degradation. Thus, not only the carbon concentration but also the nature of carbon-related defect states is an important factor in achieving electrically stable semi-insulating buffer layers. Previous theoretical and experimental studies have suggested that carbon atoms in GaN can occupy different lattice sites, such as substitutional carbon on nitrogen (C_N) or gallium (C_{Ga}) sites, interstitial carbon (C_i), or form more complex defect configurations involving multiple carbon atoms.^{16–18} Each of these configurations is associated with distinct charge transition levels and electrical activity. The formation of specific carbon-related defects is known to depend on several factors, including growth conditions, carbon concentration, and the carbon incorporation pathway.^{9,10} In metalorganic vapor phase epitaxy (MOVPE), carbon can be incorporated either unintentionally from metalorganic precursors or intentionally by introducing external hydrocarbon sources. In addition to *in situ* doping during growth, carbon incorporation in GaN can also be achieved by ion implantation, as reported in a previous study.¹⁹ In carbon autodoping, carbon is incorporated gradually from residual decomposition products of metalorganic precursors, which allows carbon-related defects to behave in a largely isolated manner even at relatively high concentrations, as demonstrated in our previous study.²⁰ In contrast, intentional carbon doping using hydrocarbon gases such as C_2H_2 provides an alternative incorporation pathway, in which carbon is supplied independently of the metalorganic group-III precursors. This difference in incorporation chemistry may influence the dominant carbon-related defect configurations formed during growth. Our research group has previously investigated carbon-related defects in GaN grown using different doping methods, including carbon autodoping and C_2H_2 gas doping, mainly focusing on optical and magnetic properties evaluated by photoluminescence (PL) and electron spin resonance (ESR).²¹ These studies provided qualitative insight into the evolution of carbon-related defects as a function of doping method and concentration. However, it remains unclear whether the conventional isolated-point-defect model, which has been widely applied to GaN:C, is still valid when carbon is introduced via an external hydrocarbon source. In particular, systematic electrical characterization using current–voltage (I–V), capacitance–voltage (C–V), and optical deep-level transient spectroscopy (ODLTS) measurements has not been sufficiently reported for GaN:C grown using C_2H_2 gas.

In this study, we investigated the electrical characteristics and deep-level states of GaN:C grown using C_2H_2 gas, with the specific aim of clarifying whether carbon-related defects can still be described as isolated deep levels or instead evolve into interacting and distributed states depending on the carbon incorporation pathway. In contrast to our previous study on autodoped GaN:C,²⁰ where carbon-related traps retained quasi-isolated behavior even at relatively high concentrations, the present work reveals a transition to interacting and distributed defect states when carbon is introduced via C_2H_2 gas. This direct comparison under otherwise

identical conditions highlights the significant role of the carbon incorporation pathway in determining the electrical stability of semi-insulating GaN buffer layers.

II. EXPERIMENTAL METHODS

Carbon-doped GaN (hereafter referred to as GaN:C) layers were grown on Si-doped n-type freestanding GaN substrates (carrier concentration $\sim 1 \times 10^{18} \text{ cm}^{-3}$) prepared by hydride vapor phase epitaxy (HVPE) using metalorganic vapor phase epitaxy (MOVPE). The threading dislocation density (TDD) of the epitaxial GaN layers was estimated to be $\sim 1 \times 10^6 \text{ cm}^{-2}$, which is a typical value for MOVPE-grown GaN on HVPE-GaN substrates. Trimethylgallium (TMGa) and ammonia (NH_3) were used as Ga and N sources, respectively, with hydrogen and nitrogen as carrier gases. The thickness of the GaN:C layer for all samples was approximately $1 \mu\text{m}$. Carbon doping was achieved by supplying acetylene (C_2H_2) gas during GaN growth. C_2H_2 was selected as the intentional carbon source because it provides a practical route for carbon incorporation under the present MOVPE growth conditions and allows control of the carbon concentration in the investigated range. The carbon concentration was controlled by adjusting the C_2H_2 gas flow rate, while the growth temperature was kept constant at 1215°C . Three different carbon concentrations of approximately $\sim 10^{17}$, $\sim 10^{18}$, and $\sim 10^{19} \text{ cm}^{-3}$ were prepared. The carbon concentrations were evaluated by secondary ion mass spectrometry (SIMS). The SIMS profiles shown in Fig. 1 represent measured values and are used as reference profiles; slight deviations from the nominal concentrations may arise from inherent measurement uncertainties in SIMS analysis, particularly at high doping levels.

For electrical characterization, circular Schottky diodes with a diameter of $300 \mu\text{m}$ were fabricated. Semi-transparent Ni (10 nm)/Au (30 nm) layers were deposited on the front side by electron-beam (EB) evaporation as Schottky contacts to allow illumination through the contact. No annealing was performed to preserve high optical transmittance and stable Schottky characteristics. The Ni/Au Schottky contact used for optical deep-level transient spectroscopy (ODLTS) measurements enabled photon irradiation of the underlying GaN layer, and reproducible DLTS signals were observed. On the backside, Al (50 nm)/Ti (20 nm)/Au (100 nm) multilayers were deposited by EB evaporation without annealing, owing to the high conductivity of the n-type GaN substrate. Current–voltage (I–V) and capacitance–voltage (C–V) measurements were performed both with and without 385 nm LED illumination. A Boonton 7200 capacitance meter operating at 1 MHz was used for the C–V measurements, while a Keithley 4200 system was used for the I–V measurements. ODLTS was performed using 385 nm LED illumination. The measurement temperature ranged from 80 to 400 K. To evaluate the effect of illumination, measurements were also performed without LED illumination under identical bias conditions. Owing to the highly insulating nature of GaN:C, capacitance changes were often negligible in the absence of illumination. For this reason, near-sub-bandgap illumination at 385 nm was employed throughout the measurements to improve signal sensitivity and measurement reliability. This wavelength is close to the GaN band edge

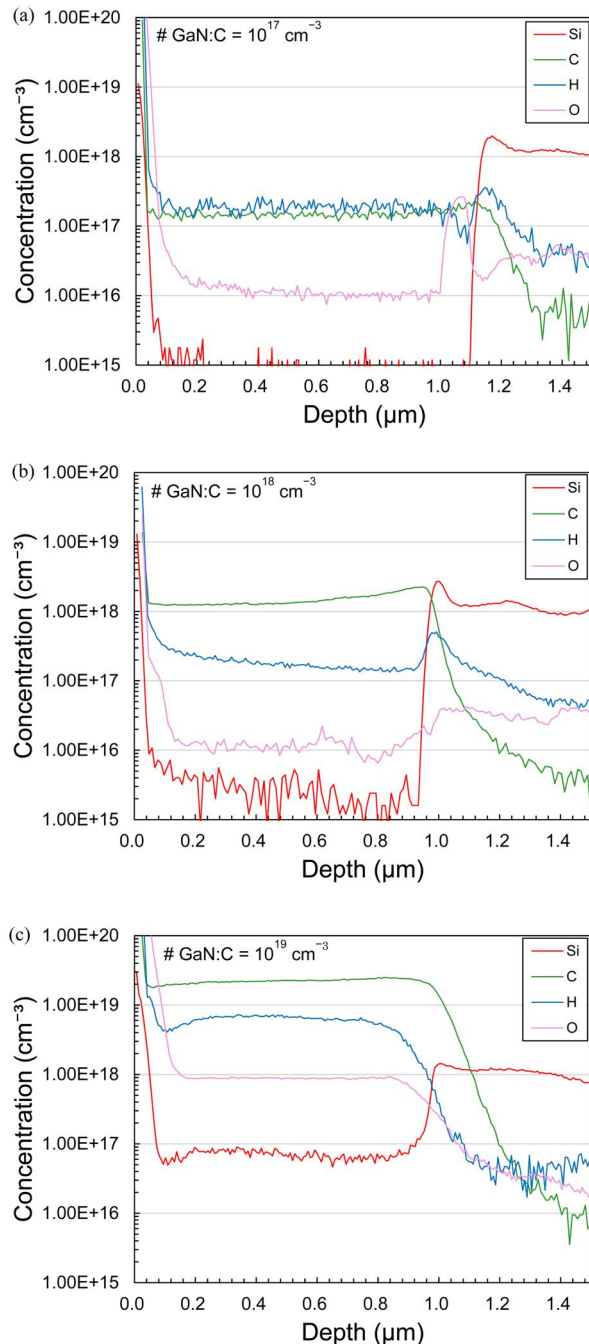


FIG. 1. Secondary ion mass spectrometry (SIMS) profiles of C_2H_2 -doped GaN:C samples with different carbon concentrations: (a) $\sim 10^{17} \text{ cm}^{-3}$, (b) $\sim 10^{18} \text{ cm}^{-3}$, and (c) $\sim 10^{19} \text{ cm}^{-3}$. The concentration profiles of Si, C, H, and O are shown as a function of depth. The SIMS data are partially adapted from our previous study.²⁰ The sample labels (C17, C18, and C19) correspond to nominal carbon concentration regimes of approximately 10^{17} , 10^{18} , and 10^{19} cm^{-3} , respectively. Adapted with permission from Honda *et al.*, *Phys. Status Solidi B* **262**, 2500012 (2025). Copyright 2025 Wiley-VCH GmbH.

and enables efficient excitation of deep levels while reducing direct band-to-band transitions. Similar sub-bandgap excitation wavelengths (in the range of 385–405 nm) have also been reported to be effective for detecting deep-level traps in GaN.^{17,18,20} Trap activation energies and capture cross sections were extracted from Arrhenius plots based on thermal emission rates derived from isothermal spectra. For clarity, temperature-dependent spectra are presented for visualization, while the quantitative analysis was based on isothermal measurements. Emission rates were obtained from the peak positions in the isothermal ODLTS spectra measured under various rate windows ($e = 1/\tau$), which were selected to resolve each trap signal across the measured temperature range. Here, T_d and T_w denote the delay time and the rate window time used in the ODLTS measurements, respectively. Reverse biases were applied depending on the C concentration: -1 V for $\sim 10^{17} \text{ cm}^{-3}$, and -4 V for $\sim 10^{18}$ and $\sim 10^{19} \text{ cm}^{-3}$. A light pulse width of 1 s was applied for the $\sim 10^{17}$ and $\sim 10^{18} \text{ cm}^{-3}$ samples, while a shorter pulse width of 10 ms was used for the $\sim 10^{19} \text{ cm}^{-3}$ sample to ensure reliable transient detection.

To minimize the influence of growth temperature on other factors, all growth parameters, including reactor pressure, V/III ratio, and precursor flow rates, were kept constant, except for the C_2H_2 gas flow rate. No additional dopants were intentionally introduced. SIMS measurements confirmed that the concentrations of other common impurities such as O, Si, and H remained nearly constant among the samples. Under identical reactor conditions, variations in intrinsic or extended defects are expected to be relatively limited. This experimental approach was adopted to focus on the influence of carbon concentration introduced by C_2H_2 doping on the electrical characteristics and trap behavior. The device structure and electrical/ODLTS measurement procedures were the same as those described in our previous study on carbon autodoped GaN:C,²⁰ except for the intentional use of C_2H_2 as an external carbon source.

III. RESULTS AND DISCUSSION

A. Electrical characteristics of C_2H_2 -doped GaN

Figure 2 shows the current–voltage (I–V) characteristics of GaN:C layers grown by intentional C_2H_2 doping with different carbon concentrations. Under dark conditions, the reverse leakage current systematically decreases with increasing carbon concentration, indicating enhanced compensation in the GaN:C layers. In contrast, the capacitance–voltage (C–V) characteristics measured without LED illumination remain nearly constant over the entire reverse-bias range for all samples as shown in Fig. 3. This result indicates that the depletion region already extends across the entire GaN:C layer, resulting in an electrically insulating state. The capacitance values measured under dark conditions are consistent with those estimated from the layer thickness, supporting the interpretation that the GaN:C layers are effectively fully depleted. Under LED illumination, however, the I–V characteristics become unstable and exhibit little dependence on carbon concentration. Under illumination, a pronounced capacitance increase is observed only for the $\sim 10^{18} \text{ cm}^{-3}$ sample, while the $\sim 10^{17}$ and $\sim 10^{19} \text{ cm}^{-3}$ samples show negligible modulation. This behavior is likely related to photoinduced carrier dynamics involving deep

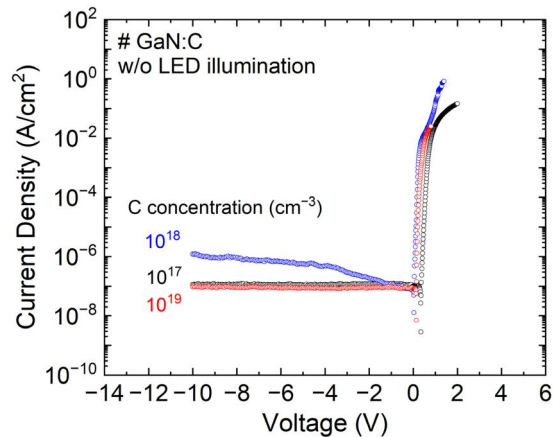


FIG. 2. Current-voltage (I-V) characteristics of C_2H_2 -doped GaN:C samples with different carbon concentrations measured under dark conditions.

defect states, particularly hole trapping, as further discussed in the ODLTS results in Sec. III C. In comparison, the weaker response in the $\sim 10^{17}$ and $\sim 10^{19} \text{ cm}^{-3}$ samples may be attributed to a lower density of active defect states ($\sim 10^{17} \text{ cm}^{-3}$) and excessive carrier trapping ($\sim 10^{19} \text{ cm}^{-3}$). These results indicate that photo-excitation influences the electrical response of C_2H_2 -doped GaN:C and that carrier emission from deep defect states plays an important role in determining the measured characteristics, as reported previously for GaN:C.¹² Notably, the capacitance response is maximized at $\sim 10^{18} \text{ cm}^{-3}$, where the balance between defect density and carrier trapping efficiency allows effective charge modulation under illumination. To further understand the transport properties associated with the observed electrical behavior, the forward I-V characteristics were analyzed to evaluate the relative resistance of the GaN:C layers. The resistance was estimated from the slope of the forward I-V curves in the high-bias region, yielding effective series resistance values of the order of 10^3 – $10^4 \Omega$. These values represent effective forward-bias series resistance, including contributions from the Schottky contact and defect-assisted transport, and, therefore, do not directly reflect the intrinsic bulk resistivity of the GaN:C layers. A non-monotonic dependence on carbon concentration was observed, with the sample at $\sim 10^{18} \text{ cm}^{-3}$ exhibiting the highest conductivity. This suggests that carrier transport cannot be explained solely by compensation, but may also involve defect-assisted conduction at intermediate concentrations. At higher carbon concentrations ($\sim 10^{19} \text{ cm}^{-3}$), the resistance increases, which may be attributed to carrier trapping or carrier localization due to a high density of defect states. These results highlight the interplay between carrier compensation and defect-assisted transport in determining the electrical properties. This trend is likely influenced by compensation effects involving multiple impurities such as O, H, and Si, which can coexist with carbon-related defects and enhance the overall compensation, thereby contributing to the observed increase in resistance. Since such series resistance may influence the dynamic response of capacitance measurements, the RC time constant was evaluated.

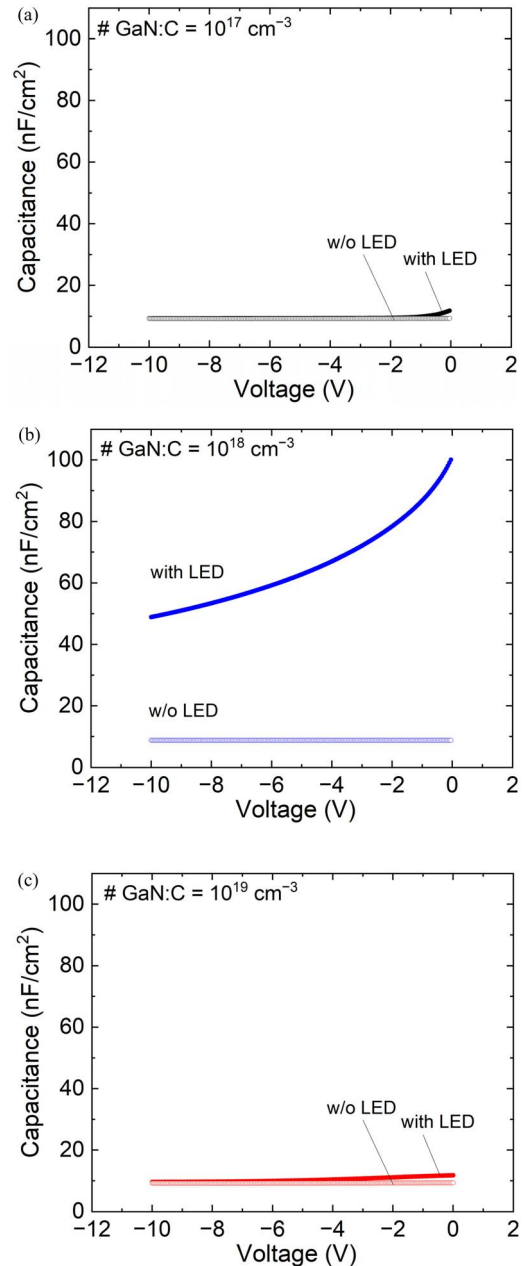


FIG. 3. Capacitance-voltage (C-V) characteristics of C_2H_2 -doped GaN:C samples with different carbon concentrations: (a) $\sim 10^{17}$, (b) $\sim 10^{18}$, and (c) $\sim 10^{19} \text{ cm}^{-3}$. Solid and open symbols represent measurements with and without 385 nm LED illumination, respectively.

Using the series resistance determined from the forward I-V characteristics and the capacitance under full depletion, the RC time constant is estimated to be comparable to or smaller than the measurement time scale (1 MHz, $\sim 1 \mu\text{s}$). Therefore, while RC-related

effects may contribute to the observed response, the full depletion is attributed to the extremely low free carrier concentration resulting from compensation by carbon-related defects. These results indicate that the observed full depletion is primarily determined by carrier compensation due to defect states, rather than by RC-related measurement limitations. Under such effectively fully depleted conditions, conventional $1/C^2$ - V analysis is not applicable because the depletion width does not vary with bias. Instead, the observed capacitance changes under illumination reflect defect-related charge dynamics rather than changes in free carrier concentration.

B. ODLTS characteristics and carbon concentration dependence

For clarity, the ODLTS features observed in this study are labeled as H1, H1', H2, H5, and so on, based on their approximate temperature ranges and activation energies. These labels are introduced for identification purposes and do not imply unique microscopic defect assignments. Figure 4 shows the ODLTS spectra of the C_2H_2 -doped GaN:C samples with increasing carbon concentration. For the lowest carbon concentration ($\sim 10^{17} \text{ cm}^{-3}$), multiple trap-related signals are observed over the measured temperature range. These peaks exhibit relatively well-defined shapes and moderate rate-window dependence, allowing conventional Arrhenius analysis to be applied. This suggests that the corresponding defect states behave in a quasi-isolated manner, i.e., defect states whose emission kinetics can still be described by a single effective Arrhenius behavior. As the carbon concentration increases to $\sim 10^{18} \text{ cm}^{-3}$, shallow ODLTS responses emerge in the $E_v + 0.15$ – 0.30 eV range. These responses show a pronounced dependence on the rate window, accompanied by apparent changes in peak position and shape, suggesting that multiple defect-related components with different emission kinetics may contribute to the observed signals. At the highest carbon concentration ($\sim 10^{19} \text{ cm}^{-3}$), the ODLTS spectra evolve into broad and distributed DLTS features without well-defined discrete peaks. The corresponding Arrhenius plots no longer yield single straight lines but instead display a distribution of apparent activation energies, as shown in Fig. 5. This result indicates a transition from quasi-isolated trap responses to interacting and distributed defect states. Such broad and distributed DLTS responses are characteristic of systems with a high density of interacting or spatially inhomogeneous defect states, where a unique Arrhenius analysis is no longer applicable.²² Table I summarizes the representative ODLTS responses observed in C_2H_2 -doped GaN:C as a function of carbon concentration, grouping the features by activation energy ranges and characteristic features. The table is intended as a phenomenological overview rather than a microscopic defect assignment, particularly at higher carbon concentrations where interacting and distributed defect states dominate.

C. Discussion: Defect-state evolution induced by C_2H_2 doping

In carbon autodoping achieved by GaN growth-temperature control, carbon atoms are incorporated gradually into the growing crystal, allowing relatively uniform distribution within the GaN

lattice. Previous studies have shown that such conditions are considered to favor the formation of predominantly isolated carbon-related point defects, often associated with interstitial carbon (C_i) at low and intermediate carbon concentrations.²⁰ As a result,

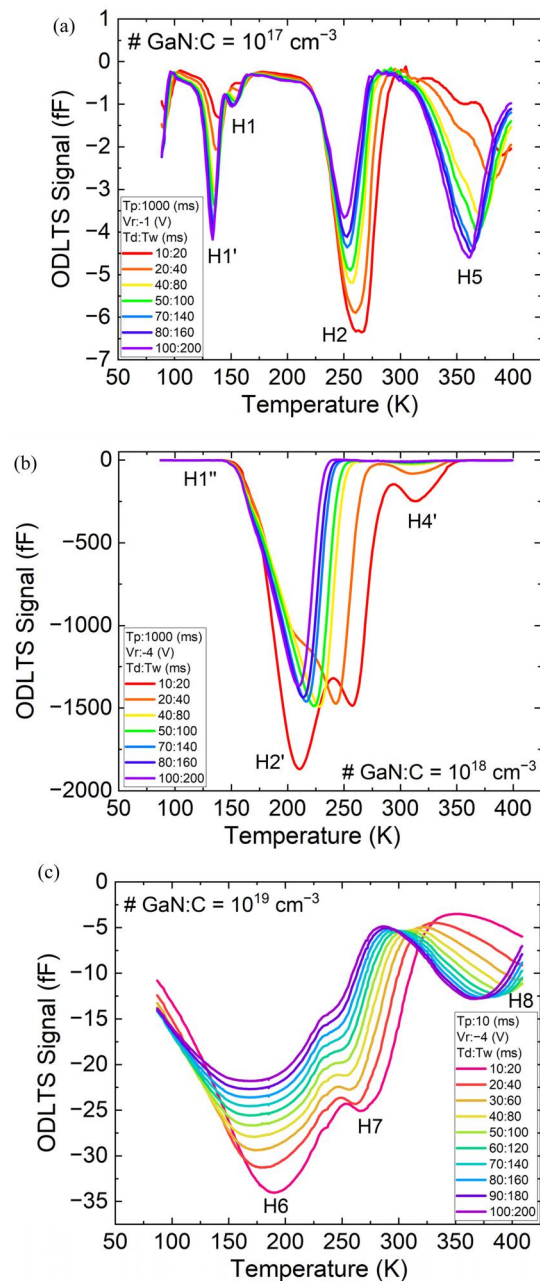


FIG. 4. Optical deep-level transient spectroscopy (ODLTS) spectra of C_2H_2 -doped GaN:C samples under 385 nm LED illumination for nominal carbon concentrations of (a) $\sim 10^{17} \text{ cm}^{-3}$ (b) $\sim 10^{18} \text{ cm}^{-3}$, and (c) $\sim 10^{19} \text{ cm}^{-3}$.

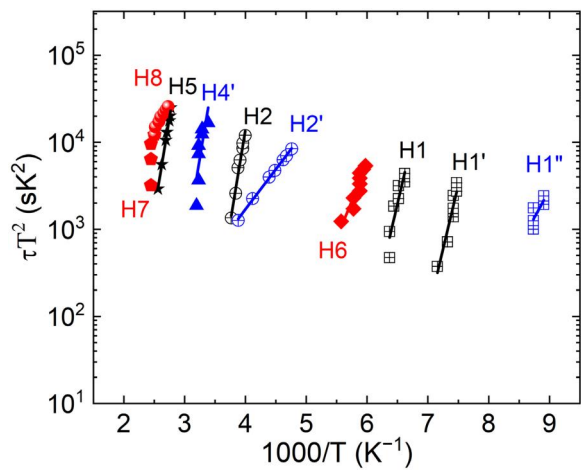


FIG. 5. Arrhenius plots (τT^2 vs $1000/T$) derived from the ODLTS spectra for nominal carbon concentrations of ($\sim 10^{17}$, $\sim 10^{18}$, and $\sim 10^{19} \text{ cm}^{-3}$) shown in Fig. 4. Deviations from a single straight-line behavior indicate that the conventional single-trap Arrhenius analysis is not applicable, particularly at higher carbon concentrations.

well-defined I–V and C–V characteristics and discrete ODLTS signals are preserved, leading to electrically stable semi-insulating behavior. In contrast, intentional C_2H_2 doping introduces a large amount of carbon from the growth surface within a short time scale, resulting in a high local carbon chemical potential near the surface. The SIMS profiles also suggest a slight extension of carbon into the underlying layer. This feature is less pronounced compared to carbon autodoping,²⁰ where carbon incorporation occurs more gradually. This difference may be attributed to the high local carbon chemical potential during C_2H_2 -based doping, which can promote diffusion or redistribution of carbon atoms. In addition, hydrogen incorporation at a level comparable to that of carbon is observed at low carbon concentration ($\sim 10^{17} \text{ cm}^{-3}$), as shown in the SIMS profiles in Fig. 1. Such incorporation is characteristic of C_2H_2 -based doping and differs from carbon autodoping,²⁰ where hydrogen incorporation is typically less significant. Hydrogen-related defects may contribute to electrical compensation through donor-like behavior, although a definitive identification is beyond the scope of the present study. From a defect perspective, in the low carbon concentration regime ($\sim 10^{17} \text{ cm}^{-3}$), C_N -related defects are generally considered to act as acceptor-like states, particularly those associated with deep levels around ~ 0.8 – 0.9 eV ,^{9,10} while donor-like defects such as C_i or hydrogen-related defects may coexist, leading to carrier compensation. These donor-like defects can act as positively charged states (e.g., interstitial hydrogen, H_i^+), whereas C_N -related defects behave as negatively charged acceptor states. As the carbon concentration increases, defect–defect interactions become increasingly important, making the distinction between donor- and acceptor-like states less clear due to the formation of interacting or distributed defect states. At higher carbon concentrations, oxygen incorporation is also observed. The oxygen is likely introduced from residual impurities

TABLE I. Summary of representative ODLTS features observed in C_2H_2 -doped GaN:C, including activation energies, capture cross sections, and possible defect origins based on previous reports.

Carbon concentration (cm^{-3})	Representative feature	Activation energy range (eV)	Capture cross section (cm^2)	Characteristic behavior	Possible origin
10^{17}	Defect peak (H_1'/H_1 -type)	~ 0.5 – 0.6	$\sim 10^{-18}$	Quasi-isolated	C_i and/or hydrogen-related defects
10^{17}	Deep peak (H_2/H_5 -type)	~ 0.8 – 0.9	$\sim 10^{-15}$	Quasi-isolated	C_N -related acceptor-like defects
10^{18} – 10^{19}	Broad shallow responses	~ 0.15 – 0.30 (e.g., 0.17 – 0.25 eV)	Not reliably determined ^a	Interacting, rate-window dependent	Shallow carbon-related complexes (possibly involving H or O)
10^{18} – 10^{19}	Broad deep response	~ 0.7 – 0.9 (e.g., 0.8 – 0.9 eV)	Not reliably determined ^a	Interacting/distributed	Distributed carbon-related defect states (possibly involving H/O complexes)

^aApparent capture cross sections vary widely due to overlapping or interacting defect states and are not considered reliable single-trap parameters.

in the reactor or background contamination during growth, as reported previously.^{16,17} Additionally, hydrogen and oxygen may form complexes with carbon atoms, as suggested by previous theoretical and experimental studies on carbon- and impurity-related defect complexes.^{9,10,23–26} These results suggest that the electrical compensation in C₂H₂-doped GaN:C arises from the interplay of donor-like and acceptor-like defect states, including hydrogen-, carbon-, and oxygen-related defects. These incorporation characteristics may also be consistent with the preferential formation of substitutional carbon on nitrogen sites (C_N) at low concentrations, as suggested by our previous ESR and PL studies.²¹ At higher concentrations, complex carbon-related configurations may form, while ESR remains sensitive to C_N-like paramagnetic components. As the average distance between carbon-related defects decreases at higher concentrations, defect–defect interactions become increasingly significant in C₂H₂-doped GaN:C. From this perspective, the defect configuration evolves with increasing carbon concentration. With increasing carbon concentration, the electrically active defect states in C₂H₂-doped GaN:C evolve from quasi-isolated defects ($\sim 10^{17}$ cm^{−3}) to interacting defect states that exhibit complex rate-dependent characteristics ($\sim 10^{18}$ cm^{−3}), and eventually to complex defect states or defect bands ($\sim 10^{19}$ cm^{−3}). In the intermediate concentration range, deep responses appearing in the E_v + 0.8–0.9 eV region appear in an energy range similar to previously reported C_N-related acceptor levels; however, the pronounced rate-window dependence and non-unique Arrhenius behavior indicate that these states cannot be described as isolated point defects, consistent with the general limitations of single-trap DLTS analysis for interacting or distributed defect states.^{27,28} Instead, they can be interpreted as reflecting interacting configurations, such as C_N in perturbed local environments or C_N–C_I-related complexes.^{23,24} At the highest concentration, the emergence of broad and distributed DLTS responses and shallow apparent states in the E_v + 0.2–0.3 eV range is qualitatively consistent with theoretical predictions for carbon-related defect configurations under high carbon incorporation.²⁵ Experimentally, such shallow dominant responses have also been observed in GaN:C at high carbon concentrations (of the order of 10²⁰ cm^{−3}), where minority carrier transient spectroscopy (MCTS) spectra are dominated by a single shallow peak around E_v + ~ 0.25 eV.²⁹ This observation is consistent with the tendency of carbon to form multi-carbon complexes (e.g., tri-carbon defects) at high carbon concentrations, which can dominate the electrical charge balance and defect-related responses.^{30,31} The formation of such interacting and complex defect states likely contributes to Fermi-level pinning^{32,33} and effectively fully depleted GaN:C layers, resulting in flat C–V characteristics under dark conditions. Under optical illumination, carriers are temporarily emitted from these defect states, causing dynamic redistribution of the internal electric field and leading to non-steady-state electrical responses in the I–V and C–V measurements. These illumination-dependent responses can be attributed to defect states identified by ODLTS, including shallow levels in the E_v + 0.15–0.30 eV range and deeper levels around E_v + 0.8–0.9 eV. In particular, the pronounced photo-response at $\sim 10^{18}$ cm^{−3} is attributed to efficient hole trapping at these defect states, while the weaker responses at $\sim 10^{17}$ and $\sim 10^{19}$ cm^{−3} reflect a limited trap density and excessive carrier trapping or localization, respectively.

Although direct microscopic identification is beyond the scope of the present study, the systematic evolution of ODLTS responses with carbon concentration supports a transition from quasi-isolated to interacting defect states.

D. Implications for GaN power device buffers

The present results indicate that the electrical properties of GaN:C are determined not only by the carbon concentration but also by the carbon incorporation pathway. Intentional C₂H₂ doping is effective in achieving carrier compensation; however, it tends to result in an effectively fully depleted state and photosensitive behavior at high carbon concentrations due to the formation of interacting and complex defect states. In contrast, carbon autodoping via growth-temperature control is expected to provide more electrically stable buffer layers under typical operating conditions by maintaining predominantly isolated defect states. These findings highlight the importance of selecting and optimizing the carbon incorporation method depending on the target device operation and reliability requirements. Under high electric field conditions in practical HEMT operations, such interacting and spatially distributed defect states may contribute to enhanced carrier trapping and transient electrical instability. In contrast, buffer layers dominated by quasi-isolated defect states are expected to exhibit more predictable compensation characteristics. Although further device-level investigation is required, the present results suggest that the carbon incorporation pathway should be carefully considered in buffer design.

IV. CONCLUSIONS

The electrical characteristics and deep-level states of GaN:C layers grown by intentional C₂H₂ doping were systematically investigated by I–V, C–V, and ODLTS measurements. The C₂H₂-doped GaN:C layers showed a resistive and effectively fully depleted state at high carbon concentrations, accompanied by pronounced photo-sensitive electrical responses. ODLTS analysis revealed an evolution of carbon-related defect states with increasing carbon concentration: quasi-isolated trap states were observed at $\sim 10^{17}$ cm^{−3}, while broad and distributed interacting defect states became dominant at $\sim 10^{19}$ cm^{−3}. At high carbon concentrations, the emergence of broad and distributed DLTS responses and shallow apparent states suggests the formation of interacting and complex carbon-related defect configurations rather than isolated point defects. These results demonstrate that the electrical properties of GaN:C are influenced not only by the carbon concentration but also by the carbon incorporation pathway. Intentional C₂H₂ doping is associated with enhanced defect–defect interactions and the formation of complex defect configurations at high concentrations, whereas carbon autodoping via growth-temperature control appears to maintain more quasi-isolated defect responses and is, therefore, expected to provide electrically more stable buffer characteristics. The present findings provide important insight into carbon-related defect engineering in GaN and highlight the necessity of optimizing the carbon incorporation method to achieve reliable and stable semi-insulating buffer layers for GaN power device applications.

ACKNOWLEDGMENTS

This work was supported by the Research Foundation for the Electrotechnology of Chubu (REFEC), Japan, and also by the Tatematsu Foundation, Japan. A portion of this research was conducted at the Next-Generation Biomaterials Hub, Nagoya University, supported by the Advanced Research Infrastructure for Materials and Nanotechnology in Japan of the Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan (Proposal No. JPMXP1226NU5203). This work was also carried out under the joint usage/research program of the Institute of Materials and Systems for Sustainability (IMaSS), Nagoya University.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Anna Honda: Conceptualization (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Writing – original draft (lead); Writing – review & editing (lead). **Momoko Inayoshi:** Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal). **Noriyuki Taoka:** Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal). **Wakana Takeuchi:** Data curation (lead); Formal analysis (equal); Investigation (equal); Visualization (lead). **Takeshi Kato:** Writing – review & editing (equal). **Hiroataka Watanabe:** Conceptualization (equal); Investigation (equal); Methodology (lead); Writing – review & editing (equal). **Yoshio Honda:** Conceptualization (equal); Project administration (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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